Solutions Tutorial 7

1. Let $f:[a,b]\to\mathbb{R}$ be a bounded function and let $P=\{x_0,x_1,x_2,\ldots,x_n\}$ be a partition of $[a,b]$. Define the upper and lower sums $U(P,f)$ and $L(P,f)$.

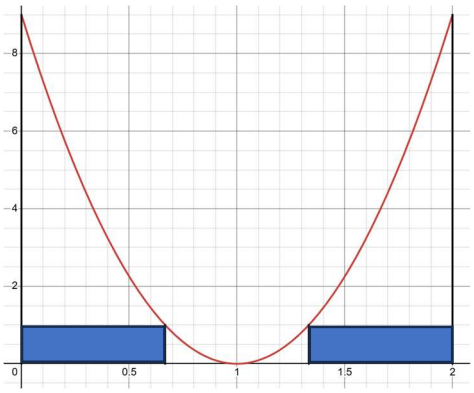
Answer. $L(P,f)=\sum_{r=0}^{k-1} I_r(f)(x_{r+1}-x_r)$, where $I_r(f)=\inf\{f(x):x\in[x_r,x_{r+1}]\}$;

$U(P,f)=\sum_{r=0}^{k-1} S_r(f)(x_{r+1}-x_r)$, where $S_r(f)=\sup\{f(x):x\in[x_r,x_{r+1}]\}$.

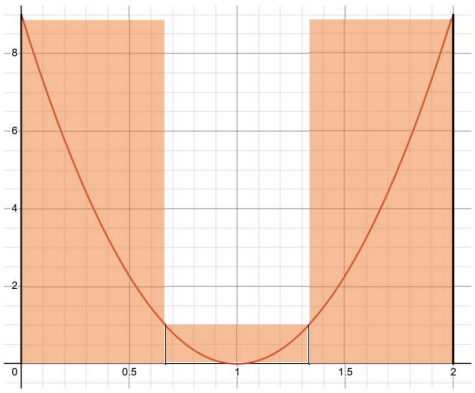
2. Compute the upper Riemann sum $U(P,f)$ and lower Riemann sum $L(P,f)$ of the function $f(x)=9(x-1)^2$ for the partition $P=\{0,\frac{2}{3},\frac{4}{3},2\}$.

Solution. $U(P,f)=9(0-1)^2\left(\frac{2}{3}-0\right)+9\left(\frac{4}{3}-1\right)^2\left(\frac{4}{3}-\frac{2}{3}\right)+9(2-1)^2\left(2-\frac{4}{3}\right)=\frac{38}{3}$.

$L(P,f)=9(\frac{2}{3}-1)^2\left(\frac{2}{3}-0\right)+0\left(\frac{4}{3}-\frac{2}{3}\right)+9(\frac{4}{3}-1)^2\left(2-\frac{4}{3}\right)=\frac{4}{3}$.



$L(P,f)$



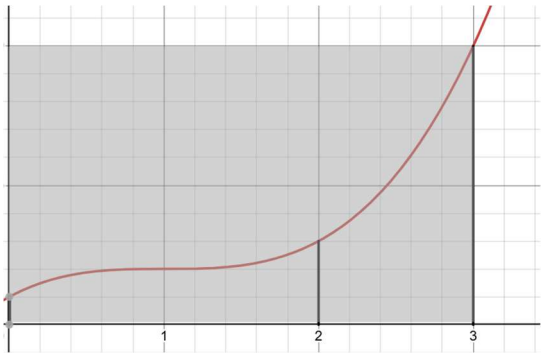
$U(P,f)$

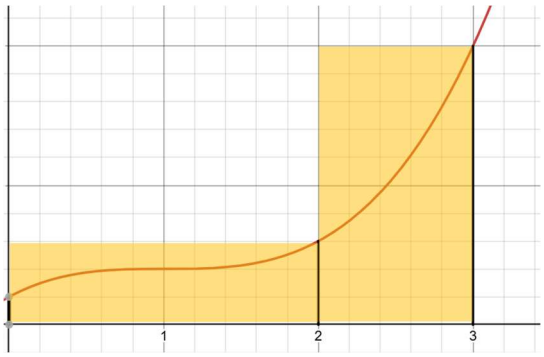
Alternative: $L(P,f)=2\times\frac{2}{3}\times 1=\frac{4}{3}$ and $U(P,f)=2\times\frac{2}{3}\times 9+\frac{2}{3}\times 1=\frac{38}{3}$.

3. Let $f:[0,3]\to\mathbb{R}$ be a bounded function. Let P and Q be partitions of $[0,3]$ given by $P=\{0,3\}$ and $Q=\{0,2,3\}$. Prove that $U(P,f)\geq U(Q,f)$.

Proof. Let $S(f)=\sup\{f(x):x\in[0,3]\}$, $S_1(f)=\sup\{f(x):x\in[0,2]\}$ and $S_2(f)=\sup\{f(x):x\in[2,3]\}$. Then $U(P,f)=3S(f)$ and $U(Q,f)=2S_1(f)+S_2(f)$. Since $S_r(f)\leq S(f)$ for $r=1,2$, it follows that $U(Q,f)\leq 2S(f)+S(f)=3S(f)=U(P,f)$.

Alternative: Let P and R be partitions of $[a,b]$ such that R is a refinement of P. Then $L(P,f)\leq L(R,f)$ and $U(R,f)\leq U(P,f)$. Here, Q is a refinement of P and hence $U(Q,f)\leq U(P,f)$.

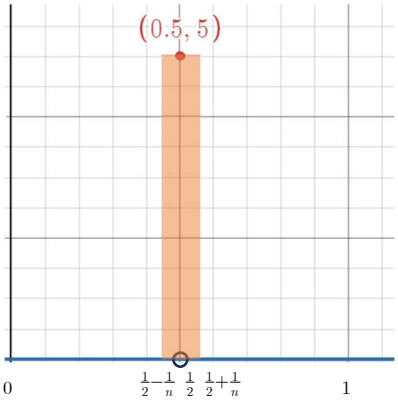




(Note that the above graphs are just for illustration, function f can be arbitrary.)

4. Let f be the function defined on $[0,1]$ by $f(x)=\begin{cases}5, & x=\frac{1}{2}\\ 0, & x\neq\frac{1}{2}\end{cases}$. Compute $U(P_n,f)$ and $L(P_n,f)$ for the partition $P_n=\left\{0,\frac{1}{2}-\frac{1}{n},\frac{1}{2}+\frac{1}{n},1\right\}, n>2$. Deduce that f is integrable on $[0,1]$ and find $\int_0^1 f(x)\,dx$.

Solution. $U(P_n,f)=0\left(\frac{1}{2}-\frac{1}{n}\right)+5\left(\frac{2}{n}\right)+0\left(1-\frac{1}{2}-\frac{1}{n}\right)=\frac{10}{n}$ and $L(P_n,f)=0$.



Since $\lim_{n\to\infty}\left(U(P_n,f)-L(P_n,f)\right)=\lim_{n\to\infty}\frac{10}{n}=0$, it follows that f is integrable on $[0,\,1]$ and

$$\begin{aligned}\int_0^1 f(x)\,dx &= \lim_{n\to\infty}\left(U(P_n,f)\right)\\ &= \lim_{n\to\infty}\left(L(P_n,f)\right)\\ &= 0.\end{aligned}$$

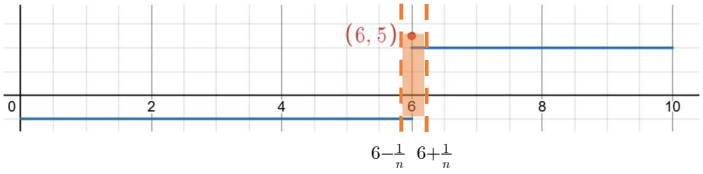
5. Let $g:[0,10]\to\mathbb{R}$ be defined by $g(x)=\begin{cases}-2, & 0\le x<6\\ 5, & x=6\\ 4, & 6<x\le 10\end{cases}$. Let P_n be the partition given by $P_n=\left\{0,6-\frac{1}{n},6+\frac{1}{n},10\right\}$. Find $U(P_n,g)$ and $L(P_n,g)$. Hence show that g is integrable on $[0,10]$ and find $\int_0^{10} g(x)\,dx$.

Solution. Compute $U(P_n,g)=-2\left(6-\frac{1}{n}\right)+5\left(6+\frac{1}{n}-\left(6-\frac{1}{n}\right)\right)+4\left(10-\left(6+\frac{1}{n}\right)\right)=4+\frac{8}{n}$ and $L(P_n,g)=-2\left(6-\frac{1}{n}\right)-2\left(6+\frac{1}{n}-\left(6-\frac{1}{n}\right)\right)+4\left(10-\left(6+\frac{1}{n}\right)\right)=4-\frac{6}{n}$. Now

$$\lim_{n\to\infty}\left(U(P_n,g)-L(P_n,g)\right)=\lim_{n\to\infty}\frac{14}{n}=0.$$

So, g is integrable on $[0,\,10]$ and $\int_0^{10} g(x)\,dx=\lim_{n\to\infty}U(P_n,g)=\lim_{n\to\infty}\left(4+\frac{8}{n}\right)=4$.

Alternative: $\lim_{n\to\infty}\left[U(P_n,g)-L(P_n,g)\right]=\lim_{n\to\infty}7(\frac{2}{n})=0$.



6. Let $f:[a,b]\to\mathbb{R}$ be increasing on $[a,b]$. Compute the upper and lower sums $U(P,f)$ and $L(P,f)$ for the partition $P=\{x_0,x_1,x_2,...,x_n\}$.

Solution. Since f is increasing, it follows that for each $r=1,\ \ldots,\ k-1$,

$$S_r(f)=\sup\{f(x):x\in[x_r,x_{r+1}]\}=f(x_{r+1})\ \text{ and }\ I_r(f)=\inf\{f(x):x\in[x_r,x_{r+1}]\}=f(x_r)$$

$$\Rightarrow U(P,f)=\sum_{r=0}^{n-1}f(x_{r+1})(x_{r+1}-x_r)\ \text{ and }\ L(P,f)=\sum_{r=0}^{n-1}f(x_r)(x_{r+1}-x_r).$$

7. Let $f:[a,b]\to\mathbb{R}$ be integrable on $[a,b]$, and $f(x)\ge m$ for all $x\in[a,b]$. Prove that for any partition P of $[a,b]$, $L(P,f)\ge m(b-a)$. Deduce that $\int_a^b f(x)\,dx\ge m(b-a)$. By considering $f(x)=\frac{1}{x}$, prove that for $a>0$, $\ln(1+\frac{1}{a})\ge\frac{1}{a+1}$.

Proof. For $x\in[a,b]$, it is given that $f(x)\ge m$, so $I_r(f)\ge m$. It then follows that

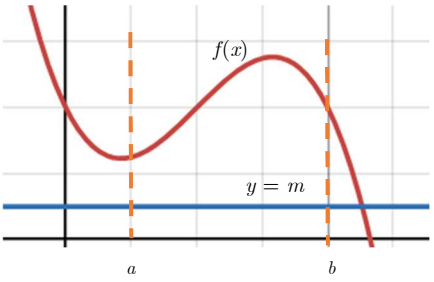
$$L(P,f)=\sum_{r=0}^{k-1}I_r(f)(x_{r+1}-x_r)\ge\sum_{r=0}^{k-1}m(x_{r+1}-x_r)=m\sum_{r=0}^{k-1}(x_{r+1}-x_r)=m(b-a),$$

$$\int_a^b f(x)\,dx=\bar{L}(f)=\sup\{L(P,f):P\ \text{is a partition of}\ [a,b]\}\ge L(P,f)\ge m(b-a).$$

(Recall: For any partitions P, Q, $L(P,f)\le\bar{L}(f)\le\underline{U}(f)\le U(Q,f)$.) Let $b=a+1$. Then for $0<a\le x\le a+1=b$, $f(x)=\frac{1}{x}\ge\frac{1}{a+1}$. Choose $m=\frac{1}{a+1}$. By the Fundamental Theorem of Calculus and the result obtained above, we conclude that

$$\int_a^b f(x)\ dx\ge m(b-a)\Rightarrow\int_a^{a+1}\tfrac{1}{x}\ dx\ge m\Rightarrow\ln(1+\tfrac{1}{a})\ge\tfrac{1}{a+1},$$

since $\int_a^{a+1}\frac{1}{x}dx=\ln(1+a)-\ln a=\ln(1+\frac{1}{a})$.



Geometrically, we see that the area under the curve $f(x)$ on interval $[a,b]$ is greater (or equal if $f(x)=m$) than the area under the line $y=m$ on interval $[a,b]$.

8. Let $g:\left[0,1\right]\to\mathbb{R}$ be a function defined by $g(x)=\frac{1}{x+1}$. Let $n\in\mathbb{N}$ and $P_n=\{0,\frac{1}{n},\frac{2}{n},...,\frac{n-1}{n},1\}$ be a partition.

(a) Show that g is Riemann integrable on $\left[0,1\right]$.

(b) Use the Lower Riemann sum in (a) to prove that

$$\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{2n-1}+\frac{1}{2n}\right)=\ln 2\,.$$

Solution. (a) $U(P_n,g)=\frac{1}{n}\left(\frac{1}{0+1}+\frac{1}{\frac{1}{n}+1}+...+\frac{1}{\frac{n-1}{n}+1}\right)$, $L(P_n,g)=\frac{1}{n}\left(\frac{1}{\frac{1}{n}+1}+\frac{1}{\frac{2}{n}+1}...+\frac{1}{\frac{n-1}{n}+1}+\frac{1}{1+1}\right)$.

$$U(P_n,g)-L(P_n,g)=\frac{1}{n}\left(1-\frac{1}{2}\right)=\frac{1}{2n}\Rightarrow\lim_{n\to\infty}\left(U(P_n,g)-L(P_n,g)\right)=0$$

and so g is integrable on $[0,1]$.

(b) $L(P_n,g)=\frac{1}{n}\left(\frac{1}{\frac{1}{n}+1}+\frac{1}{\frac{2}{n}+1}...+\frac{1}{\frac{n-1}{n}+1}+\frac{1}{1+1}\right)=\sum_{r=1}^{n}\frac{1}{n+r}$. Since g is integrable on $[0,\ 1]$, it follows that $\lim_{n\to\infty}\left(\sum_{r=1}^{n}\frac{1}{n+r}\right)=\int_0^1\frac{1}{x+1}\,dx=\left[\ln(x+1)\right]_0^1=\ln 2\,.$

9. (a) State the Fundamental Theorem of Calculus.

(b) Let $F(x)=\int_x^{\sin x}\frac{1}{2+t^2}\,dt$. Explain why F is differentiable and compute $F'(x)$.

Proof. (a) Fundamental Theorem of Calculus (FTOC)

Let $f,g:[a,b]\to\mathbb{R}$ be integrable functions on $[a,b]$.

(i) Suppose $F:[a,b]\to\mathbb{R}$ is a function that satisfies $F'(x)=f(x)\ \forall x\in[a,b]$. Then

$$\int_a^b f(x)\,dx=F(b)-F(a)\,.$$

(ii) Let $G(x):=\int_a^x g(t)\,dt\ \forall x\in[a,b]$. Then G is continuous on $[a,b]$. If g is continuous at a point $c\in[a,b]$, then G is differentiable at c and $G'(c)=g(c)$.

(b) By FTOC(ii), if $g(t)$ is a continuous function on a closed interval $[a,b]$, then $G(x):=\int_a^x g(t)\,dt$ is differentiable for all $x\in[a,b]$ and $G'(x)=g(x)$. Let $g(t)=\frac{1}{2+t^2}$.

Then g is continuous everywhere since $2+t^2\neq 0\ \forall t\in\mathbb{R}$. Moreover,

$$F(x)=\int_x^{\sin x}\frac{1}{2+t^2}\,dt=\int_a^{\sin x}\frac{1}{2+t^2}\,dt-\int_a^x\frac{1}{2+t^2}\,dt=G(\sin x)-G(x)$$
$$\Rightarrow F'(x)=(\cos x)\ G'(\sin x)-G'(x)=\frac{\cos x}{2+\sin^2 x}-\frac{1}{2+x^2}$$

Also, since G is differentiable, F is differentiable.

Extra Exercises

10. Evaluate the following sums.

(a) $\displaystyle\sum_{i=0}^{100}\left(\frac{3}{5}\right)^i$ (b) $\displaystyle\sum_{i=50}^{100}\left(\frac{3}{5}\right)^i$ (c) $\displaystyle\sum_{i=1}^{10}(i^2-3i+5)$ (d) $\displaystyle\sum_{n=1}^{b}\left(\frac{1}{e^n}+en^3\right),\ b\in\mathbb{Z}^+\setminus\{1\}$.

Solution. $\displaystyle\sum_{i=0}^{n-1}\alpha^i=\frac{1-\alpha^n}{1-\alpha},\ \alpha\in\mathbb{R}\setminus\{0,1\}$, $\displaystyle\sum_{i=1}^{n}i=\frac{n(n+1)}{2}$, $\displaystyle\sum_{i=1}^{n}i^2=\frac{n(n+1)(2n+1)}{6}$, $\displaystyle\sum_{i=1}^{n}i^3=\left(\sum_{i=1}^{n}i\right)^2$.

(a) $\displaystyle\sum_{i=0}^{100}\left(\frac{3}{5}\right)^i=\frac{1-\left(\frac{3}{5}\right)^{101}}{1-\frac{3}{5}}=\frac{5}{2}\left(1-\left(\frac{3}{5}\right)^{101}\right)$

(b) $\displaystyle\sum_{i=50}^{100}\left(\frac{3}{5}\right)^i=\sum_{i=0}^{100}\left(\frac{3}{5}\right)^i-\sum_{i=0}^{49}\left(\frac{3}{5}\right)^i=\frac{5}{2}\left(1-\left(\frac{3}{5}\right)^{101}\right)-\frac{5}{2}\left(1-\left(\frac{3}{5}\right)^{50}\right)=\frac{5}{2}\left(\frac{3}{5}\right)^{50}\left(1-\left(\frac{3}{5}\right)^{51}\right)$

(c) $\displaystyle\sum_{i=1}^{10}(i^2-3i+5)=\sum_{i=1}^{10}i^2-3\sum_{i=1}^{10}i+5\sum_{i=1}^{10}1=\frac{10(11)(21)}{6}-\frac{3(10)(11)}{2}+5(10)=270$

(d) $\displaystyle\sum_{n=1}^{b}\left(\frac{1}{e^n}+en^3\right)=\frac{1}{e}\cdot\frac{1-1/e^b}{1-1/e}+e\frac{b^2(b+1)^2}{4}=\frac{1}{e^b}\left(\frac{e^b-1}{e-1}\right)+\frac{e}{4}b^2(b+1)^2,\ b\in\mathbb{Z}^+\setminus\{1\}$.

For Question 11 to 14. Note that the given functions for the following questions are all integrable on given intervals.

11. If $\displaystyle\int_0^b\cos x\ dx=\sin b$, then what is $\displaystyle\int_a^b\cos x\ dx$?

Solution.
$$\begin{aligned}\int_a^b\cos x\ dx&=\int_a^0\cos x\ dx+\int_0^b\cos x\ dx\\&=-\int_0^a\cos x\ dx+\int_0^b\cos x\ dx\\&=-\sin a+\sin b\\&=\sin b-\sin a\end{aligned}$$

12. The functions $f(x)$ and $g(x)$ satisfy

$$\int_0^{-1}f(x)\ dx=1\qquad\int_0^2f(x)\ dx=2\qquad\int_{-1}^0g(x)\ dx=3\qquad\int_0^2g(x)\ dx=4$$

Find $\displaystyle\int_{-1}^2\Big[3g(x)-f(x)\Big]dx$.

Solution.
$$\begin{aligned}\int_{-1}^2\Big[3g(x)-f(x)\Big]dx&=3\int_{-1}^2g(x)\ dx-\int_{-1}^2f(x)\ dx\\&=3\left[\int_{-1}^0g(x)\ dx+\int_0^2g(x)\ dx\right]-\left[\int_{-1}^0f(x)\ dx+\int_0^2f(x)\ dx\right]\\&=3\Big[3+4\Big]-\left[-\int_0^{-1}f(x)\ dx+\int_0^2f(x)\ dx\right]\\&=3\Big(7\Big)-(-1+2)=20\end{aligned}$$

13. Let $f(x)=\frac{\sin x}{\ln(3+x^2)}$. Evaluate $\int_{-2012}^{2012} f(x)\,dx$.

Solution. $f(x)=\frac{\sin x}{\ln(3+x^2)}=-f(-x)$ is an odd function. So, $\int_{-2012}^{2012}\frac{\sin x}{\ln(3+x^2)}dx=0$.

14. Is the derivative of an even function even or odd? What about the odd function?

Solution. Suppose $f(x)$ is an odd function. Then $f(-x)=-f(x)$. Differentiate on both sides to get $-f'(-x)=-f'(x)\Rightarrow f'(-x)=f'(x)$. Hence, $f'(x)$ is even (meaning to say the derivative of an odd function is even).

Suppose $f(x)$ is even. Then $f(-x)=f(x)$. Differentiate on both sides to obtain $f'(-x)=f'(x)\Rightarrow -f'(-x)=f'(x)$. Hence, $f'(x)$ is odd (meaning to say the derivative of an even function is odd).

15. For the function $f(x)=x^3-\sin 2x$, find its antiderivative $F(x)$ that satisfies $F(0)=1$.

Solution. Note that $F(x)=\frac{1}{4}x^4+\frac{1}{2}\cos 2x+C$. Given $F(0)=1$. So, $C=\frac{1}{2}$.

16. Evaluate $\int_{-1}^{2}\left|2x\right|\,dx$.

Solution. Since $\int_{-1}^{2}\left|2x\right|\,dx=\int_{-1}^{0}(-2x)\,dx+\int_{0}^{2}2x\,dx$, we will find the answers of two integrations, i.e., $\int_{-1}^{0}(-2x)\,dx=\int_{0}^{1}2x\,dx=\lim_{n\to\infty}\sum_{i=1}^{n}\frac{1}{n}\cdot\frac{2i}{n}=\lim_{n\to\infty}\frac{2}{n^2}\frac{n(n+1)}{2}=1$

(using the partition $\{0,\frac{1}{n},\frac{2}{n},...,\frac{n-1}{n},1\}$ to find the upper Riemann sum)

$$\int_{0}^{2}2x\,dx=\lim_{n\to\infty}\sum_{i=1}^{n}\frac{2}{n}\cdot\frac{4i}{n}=\lim_{n\to\infty}\frac{8}{n^2}\frac{n(n+1)}{2}=4$$

(using the partition $\{0,\frac{2}{n},\frac{4}{n},...,\frac{2(n-1)}{n},2\}$ to find the upper Riemann sum)

Hence, $\int_{-1}^{2}\left|2x\right|\,dx=5$.

Alternative 1: Find the area of two triangles (geometrically) to obtain the answer.

Alternative 2: Since f is integrable on $\left[-1,2\right]$, by the Fundamental Theorem of Calculus, $\int_{-1}^{2}\left|2x\right|\,dx=\int_{-1}^{0}(-2x)\,dx+\int_{0}^{2}2x\,dx=G(0)-G(-1)+F(2)-F(0)$, where G and F are antiderivatives, i.e., $G'(x)=-2x$ and $F'(x)=2x$.

17. Evaluate the integral by interpreting it as a signed area using geometry: $\int_0^5 x\ dx$.

Solution. Check that (geometrically) the area of the triangle is $\frac{1}{2}(5^2)=\frac{25}{2}$ unit^2. We can use the lower or upper Riemann sum to verify the answer. Let L be the lower Riemann sum and U be the upper Riemann sum.

Partition $[0,5]$ into n equal-length subintervals, i.e., $P=\left\{0,\frac{5}{n},\frac{10}{n},...,5-\frac{5}{n},5\right\}$.

$$L=\frac{5}{n}\sum_{i=0}^{n-1}\frac{5i}{n}=\frac{25}{n^2}\left(\frac{n(n-1)}{2}\right)=\frac{25(n-1)}{2n}\to\frac{25}{2}=\overline{L}\ \text{ when }\ n\to\infty.$$

$$U=\frac{5}{n}\sum_{i=1}^{n}\frac{5i}{n}=\frac{25}{n^2}\left(\frac{n(n+1)}{2}\right)=\frac{25(n+1)}{2n}\to\frac{25}{2}=\underline{U}\ \text{ when }\ n\to\infty.$$

Then $\int_0^5 x\ dx=\overline{L}=\underline{U}=\frac{25}{2}$.

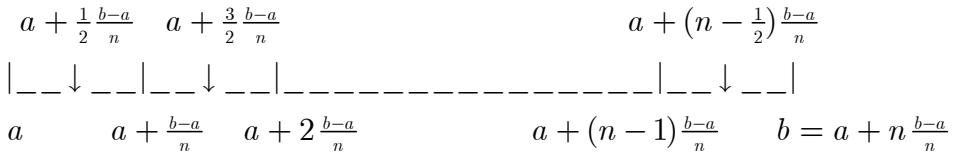
Let f be an integrable function on interval $[a,b]$. We partition $[a,b]$ into n equal-length subintervals, $P=\left\{a,a+\frac{b-a}{n},a+2\left(\frac{b-a}{n}\right),...,a+(n-1)\left(\frac{b-a}{n}\right),a+n\left(\frac{b-a}{n}\right)\right\}$.

Define the left, right, midpoint Riemann sum on $[a,b]$, as

$L=\sum_{i=1}^{n}f(c_i)\frac{b-a}{n}$, where $c_i=a+(i-1)\frac{b-a}{n}$.

$R=\sum_{i=1}^{n}f(c_i)\frac{b-a}{n}$, where $c_i=a+i\frac{b-a}{n}$.

$M=\sum_{i=1}^{n}f(c_i)\frac{b-a}{n}$, where $c_i=a+(i-\frac{1}{2})\frac{b-a}{n}$.



Note: 1. L is used as Left Riemann sum (same symbol as the lower Riemann sum).

2. L, R and M are approximations to $\int_a^b f$ if f is integrable on $[a,b]$.

3. Theorem. Let $P=\{a=x_0,x_1,...,x_{n-1},x_n=b\}$. If f is integrable on $[a,b]$, then for $c_i\in[x_i,x_{i+1}]$, $\int_a^b f=\lim_{n\to\infty}\sum_{i=1}^{n}f(c_i)(x_{i+1}-x_i)=\lim_{n\to\infty}L=\lim_{n\to\infty}R=\lim_{n\to\infty}M$.

18. Given the left Riemann sum, $L=\sum_{k=1}^{4} f(1+k)\cdot 1$. Find a, b and n based on

$$L=\sum_{i=1}^{n} f(c_i)\tfrac{b-a}{n}, \text{ where } c_i=a+(i-1)\tfrac{b-a}{n}.$$

Solution. It is obvious that $n=4$. So, $\frac{b-a}{4}=1\Rightarrow b-a=4$. Thus,

$1+k\equiv a+(k-1)\frac{b-a}{n}=a+(k-1)\cdot 1=(a-1)+k\Rightarrow a=2$ and therefore $b=6$.

19. Use sigma notation to write the midpoint Riemann sum for $f(x)=x^8$ on $[5, 15]$ with $n=50$. Do not evaluate the Riemann sum.

Solution. Given $M=\sum_{i=1}^{n} f(c_i)\tfrac{b-a}{n}$, where $c_i=a+(i-\frac{1}{2})\frac{b-a}{n}$. So, $c_i=5+\frac{1}{5}(i-\frac{1}{2})$ and

$$M=\tfrac{1}{5}\sum_{i=1}^{50} f(c_i)=\tfrac{1}{5}\sum_{i=1}^{50}\Big(5+\tfrac{1}{5}(i-\tfrac{1}{2})\Big)^8.$$

Use calculator or EXCEL, or $\sum_{k=1}^{n} k^8=\frac{1}{90}(10n^9+45n^8+60n^7-42n^5+20n^3-3n)$. You may find more formula in following link: **https://mathworld.wolfram.com/PowerSum.html**

20. Estimate $\int_{-1}^{5} x^3\,dx$ using three approximating rectangles and lefthand end points.

Solution. Let $n=3$. We have the width$=\frac{5+1}{3}=2$ and $P=\{-1,1,3,5\}$.

$$L=2\Big(f(-1)+f(1)+f(3)\Big)=2\Big(-1+1+27\Big)=54\approx\int_{-1}^{5} x^3\,dx.$$

21. The value of the following limit of a right Riemann sum is equal to the area below a graph of $y=f(x)$, integrated over the interval $[0,b]$: $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{4}{n}\Big(\sin\Big(2+\frac{4i}{n}\Big)\Big)^2$. Find $f(x)$ and b.

Solution. $a=0$, $\frac{b-a}{n}=\frac{b}{n}=\frac{4}{n}\Rightarrow b=4$ and $f(c_i)=\sin^2\Big(2+\frac{4i}{n}\Big)$, where

$$c_i=a+i\tfrac{b-a}{n}=0+i\tfrac{4}{n}\ \text{(right Riemann sum)}.$$

So, $f(c_i)=\sin^2\Big(2+c_i\Big)\Rightarrow f(x)=\sin^2(2+x)$.

22. Let $R_n = \sum_{i=1}^{n} \frac{ie^{i/n}}{n^2}$. Express $\lim_{n\to\infty} R_n$ as a definite integral but not evaluate it.

Solution. $R_n = \sum_{i=1}^{n} \frac{ie^{i/n}}{n} \cdot \frac{1}{n} = \sum_{i=1}^{n} f(c_i) \cdot \frac{1}{n}$, where $c_i = 0 + i\frac{1}{n}$. Thus, $f(x) = xe^x$. Since f is continuous on $[0,1]$, it follows that f is integrable on $[0,1]$. Hence, $\lim_{n\to\infty} R_n = \int_0^1 xe^x dx$.

23. True or false? Provide a brief justification.

(a) If $f(x)$ is continuous on $[1,\pi]$ and differentiable on $(1,\pi)$, then

$$\int_1^{\pi} f'(x)\ dx = f(\pi) - f(1).$$

(b) $\int_{-1}^{1} \frac{dx}{x^2} = 0$. (c) If f is continuous on $[a,b]$ then $\int_a^b xf(x)\ dx = x\int_a^b f(x)\ dx$.

Solution. (a) True, by part 2 of the Fundamental Theorem of Calculus.

(b) False. $f(x) = \frac{1}{x^2}$ is undefined at $x = 0$ and unbounded on $[-1,1]$. Since $f(x)$ is an even function, it follows that $\int_{-1}^{1} \frac{dx}{x^2} = 2\int_0^1 \frac{dx}{x^2} = 2 \times \lim_{a\to 0^+} \int_a^1 \frac{dx}{x^2} = 2 \times \lim_{a\to 0^+} \left[\frac{1}{x}\right]_1^a = \infty$. Thus, the improper integration does not exist.

(c) False. Provide any counter example, for example $f(x) = \frac{1}{x}$.

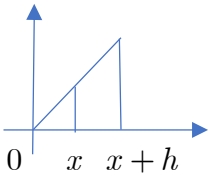
24. Suppose $F(x) = \int_2^x e^{1/t} dt$. Find the slope of $y = F(x)$ at $x = 3$.

Solution. By FTOC, $F'(x) = e^{1/t}$. So, the slope of $y = F(x)$ at $x = 3$ is $F'(3) = e^{1/3}$.

25. Let $F(x) = \int_a^x t\ dt$. Using the definition of the derivative, find $F'(x)$.

Solution. $F'(x) = \lim_{h\to 0} \frac{F(x+h)-F(x)}{h} = \lim_{h\to 0} \frac{1}{h}\left(\int_a^{x+h} t\ dt - \int_a^x t\ dt\right) = \lim_{h\to 0} \frac{1}{h}\left(\int_x^{x+h} t\ dt\right)$.

Note that $\int_x^{x+h} t\ dt = \frac{1}{2}h(x + x + h)$ (by the Riemann sum, or area of trapezium).



Hence, $F'(x) = \lim_{h\to 0} \frac{h(2x+h)}{2h} = \lim_{h\to 0} \frac{2x+h}{2} = x$.

26. Give a continuous function $f(x)$ so that $F(x) = \int_0^x f(t)\ dt$ is a constant.

Answer. By FTOC, $F'(x) = f(x)$. Choose $f(x) = 0$. Then $F(x) = C =$ constant.

27. For the following functions, evaluate the integral $\int_0^1 f(x)\,dx$ using Riemann sums, and the partitions: $P_n = \left\{0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, ..., \frac{n-1}{n}, 1\right\}$, $n \in \mathbb{N}$.

(a) $f(x) = x$

(b) $f(x) = x^2$

Also, work out these integrals using the fundamental theorem of calculus.

Solution. Using the partition suggested in the question:

(a) $L(P_n, f) = \sum_{r=0}^{n-1} I_r(f)(x_{r+1} - x_r) = \sum_{r=0}^{n-1} \frac{r}{n}\frac{1}{n} = \frac{1}{n^2}\sum_{r=0}^{n-1} r = \frac{1}{n^2}\frac{n(n-1)}{2} \to \frac{1}{2}$ as $n \to \infty$.

$U(P_n, f) = \sum_{r=0}^{n-1} S_r(f)(x_{r+1} - x_r) = \sum_{r=0}^{n-1} \frac{r+1}{n}\frac{1}{n} = \frac{1}{n^2}\sum_{r=0}^{n-1}(r+1) = \frac{1}{n^2}\frac{n(n+1)}{2} \to \frac{1}{2}$ as $n \to \infty$.

Since $\lim_{n\to\infty} U(P_n, f) = \lim_{n\to\infty} L(P_n, f) = \frac{1}{2}$, $f(x) = x$ is integrable (because it is continuous everywhere) and $\int_0^1 f(x)\,dx = \frac{1}{2}$. By FTOC, $\int_0^1 f(x)\,dx = \int_0^1 x\,dx = \frac{1}{2}x\Big|_0^1 = \frac{1}{2}$.

(b) Suppose $f(x) = x^2$. Let $P_n = \{0, \frac{1}{n}, ..., \frac{n}{n}\}, n \in \mathbb{N}$. Then

$$L(P_n, f) = \frac{1}{n}\left(\left(\frac{0}{n}\right)^2 + \left(\frac{1}{n}\right)^2 + ... + \left(\frac{n-1}{n}\right)^2\right) = \frac{1}{n^3}\frac{n-1}{6}(n)(2n-1) = \frac{1}{6n^2}(n-1)(2n-1)$$
$$U(P_n, f) = \frac{1}{n}\left(\left(\frac{1}{n}\right)^2 + \left(\frac{2}{n}\right)^2 + ... + \left(\frac{n}{n}\right)^2\right) = \frac{1}{n^3}\frac{n}{6}(n+1)(2n+1) = \frac{1}{6n^2}(n+1)(2n+1)$$

where $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$. Let

$$\overline{L}(f) = \sup\{L(P_n, f) : P_n \text{ is a partition of } [0,\, 1]\},$$
$$\underline{U}(f) = \inf\{U(P_n, f) : P_n \text{ is a partition of } [0,\, 1]\}.$$

Then

$$\overline{L} = \lim_{n\to\infty} \frac{1}{6n^2}(n-1)(2n-1) = \frac{1}{3} = \lim_{n\to\infty} \frac{1}{6n^2}(n+1)(2n+1) = \underline{U}.$$

So, f is integrable on $[0,\, 1]$ and $\int_0^1 f dx = \frac{1}{3}$.

By FTOC, $\int_0^1 f(x)\,dx = \int_0^1 x^2\,dx = \frac{1}{3}x^3\Big|_0^1 = \frac{1}{3}$.

28. Let function $G:[-1,1]\to\mathbb{R}$ defined by $G(x):=\int_{-1}^{x} g(t)\,dt\,,\ x\in[-1,1]$.

(a) Consider the function $g:[-1,1]\to\mathbb{R}$ defined by $g:=\begin{cases}0 & -1\le x<0\\ 1 & 0\le x<1\end{cases}$. Calculate the values of the function G and show that G is continuous on $[-1,1]$, but is not differentiable on $(-1,1)$.

(b) Consider the function $g:[-1,1]\to\mathbb{R}$ defined by $g(x):=\begin{cases}0 & x\neq 0\\ 1 & x=0\end{cases}$. Show that in this case G is differentiable on $(-1,1)$, but $G'\neq g$.

Remark: These examples show that the continuity conditions in the Fundamental Theorem of Calculus are necessary.

Proof. (a) $G(x)=\begin{cases}\int_{-1}^{x} 0\ dt, & -1\le x<0\\ \int_{-1}^{0} 0\ dt+\int_{0}^{x} 1\ dt, & 0\le x<1\end{cases}$
$=\begin{cases}0, & -1\le x<0\\ x, & 0\le x<1\end{cases}$

Verify that G is continuous everywhere but not differentiable at $x=0$.

(b) $G(x)=\begin{cases}\int_{-1}^{x} 0\ dt, & -1\le x<0\\ \int_{-1}^{0} 0\ dt+\int_{0}^{x} 0\ dt, & 0\le x<1\end{cases}$.

So, $G(x)=0\ \forall x\in[-1,1]$.

In this case G is differentiable on $(-1,1)$ but $G'\neq g$.

Past test/final exam questions

29. (a) State the fundamental theorem of calculus (for both statements).

(b) Let $f:\mathbb{R}\to\mathbb{R}$ be a continuous function. Find $f(x)$ if

$$x^2 = 1 + \int_1^x f(t)\,dt\,, \text{ where } x \geq 1\,.$$

Solution. (a) FTOC: (i) If $f:[a,b]\to\mathbb{R}$ is integrable on $[a,b]$, and $F:[a,b]\to\mathbb{R}$ satisfies $F'(x)=f(x)$ for all $x\in[a,b]$, then $\boxed{\int_a^b f(x)\,dx = F(b)-F(a)}$.

(ii) Let $g:[a,b]\to\mathbb{R}$ be integrable on $[a,b]$ and define $G(x)=\int_a^x g(t)\,dt$ for all $x\in[a,b]$. Then, G is continuous on $[a,b]$. If g is continuous at some point $c\in[a,b]$, then G is differentiable at c and $\boxed{G'(c)=g(c)}$.

(b) By the FTOC: Let $F(x)=\int_1^x f(t)\,dt \Rightarrow F'(x)=f(x)$. Differentiate on both sides

$$x^2 = 1 + \int_1^x f(t)\,dt \Rightarrow 2x = f(x)$$

Alternative: $x^2-1=\int_1^x f(t)\,dt = F(x)-F(1) \Rightarrow 2x = f(x) \because \frac{d}{dx}F(1)=0$.

30. Consider a function $f:[0,1]\to\mathbb{R}$ and a partition P of $[0,1]$.

(a) Define what it means for f to be Riemann integrable on $[0,1]$.

(b) Use (a) to find $\int_0^1 f$ if $f(x)=x$.

Solution. (a) $f:[0,1]\to\mathbb{R}$ is integrable if there exists a partition P_n of $[0,1]$ such that

$$\lim_{n\to\infty}\left[U(P_n f)-L(P_n,f)\right]=0\,,$$

where $U(P_n,f)$ and $L(P_n,f)$ are upper and lower sum for P_n.

(b) Since f is increasing, it follows that

$$U(P_n,f)=\tfrac{1}{n}\sum_{r=0}^{n-1}\tfrac{r+1}{n}=\tfrac{1}{n^2}\sum_{r=0}^{n-1}(r+1)=\tfrac{1}{n^2}\times\tfrac{n(n+1)}{2}\to\tfrac{1}{2}\text{ as } n\to\infty\,.$$

Hence, $\lim_{n\to\infty}U(P_n f)=0=\frac{1}{2}$.

31. Prove that $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{2n}\frac{n}{4n^2+k^2}=\frac{\pi}{8}$.

Proof. Let $f:[0,1]\to\mathbb{R}$ be defined as $f(x)=\frac{1}{1+x^2}$. Let $P_n=\{0,\frac{1}{2n},\frac{2}{2n},...,\frac{2n-1}{2n},1\}$ be a partition of $[0,1]$. Then the upper and lower Riemann sums for f are

$$U(P_n,f)=\frac{1}{2n}\left[\frac{1}{1+0^2}+\frac{1}{1+(\frac{1}{2n})^2}+...+\frac{1}{1+(\frac{2n-1}{2n})^2}\right],\ L(P_n,f)=\frac{1}{2n}\left[\frac{1}{1+(\frac{1}{2n})^2}+...+\frac{1}{1+(\frac{2n-1}{2n})^2}+\frac{1}{2}\right]$$
$$\Rightarrow U-L=\frac{1}{4n}\Rightarrow\lim_{n\to\infty}(U-L)=0$$

So, f is integrable on $[0,1]$. By FTOC $\lim\limits_{n\to\infty}L=\int_0^1 f\,dx$:

$$\frac{\pi}{4}=\tan^{-1}x\Big|_0^1=\int_0^1\frac{1}{1+x^2}\,dx=\lim_{n\to\infty}\frac{1}{2n}\left[\frac{1}{1+(\frac{1}{2n})^2}+...+\frac{1}{1+(\frac{2n-1}{2n})^2}+\frac{1}{2}\right]=\lim_{n\to\infty}\sum_{k=1}^{2n}\frac{1}{2n+\frac{k^2}{2n}}=\lim_{n\to\infty}\sum_{k=1}^{2n}\frac{2n}{4n^2+k^2}$$

which implies that $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{2n}\frac{n}{4n^2+k^2}=\frac{\pi}{8}$.

Verify:	From	change to
1.	dx	$\frac{1}{2n}$
2.	x	$\frac{k}{2n}$
3.	$k=0$ to $2n-1$; $k=1$ to $2n$	$x=0$ to 1
4.	$\lim\limits_{n\to\infty}\sum$	$\int$
5.	$\frac{1}{2n}\lim\limits_{n\to\infty}\sum\limits_{k=1}^{2n}\frac{1}{1+(\frac{k}{2n})^2}$	$\int_{x=0}^{x=1}\frac{1}{1+x^2}\,dx$

32. Find $\lim\limits_{n\to\infty}\frac{1}{n^{2026}}\sum\limits_{k=1}^{n}k^{2025}$. Hence, deduce $\lim\limits_{n\to\infty}\frac{1}{n^{N+1}}\sum\limits_{k=1}^{n}k^N$ where $N\in\mathbb{N}$ is a fixed number.

Solution. Let $f:[0,1]\to\mathbb{R}$ be defined as $f(x)=x^{2025}$. Let $P_n=\{0,\frac{1}{n},\frac{2}{n},...,\frac{n-1}{n},1\}$ be a partition of $[0,1]$. Then the lower and upper Riemann sums for f are

$$L(P_n,f)=\frac{1}{n}\sum_{k=0}^{n-1}\left(\frac{k}{n}\right)^{2025},\ U(P_n,f)=\frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^{2025}$$
$$\Rightarrow U-L=\frac{1}{n}\Rightarrow\lim_{n\to\infty}(U-L)=0$$

So, f is integrable on $[0,1]$. By FTOC and take $U(P_n,f)$,

$$\lim_{n\to\infty}\frac{1}{n^{2026}}\sum_{k=1}^{n}k^{2025}=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}(\tfrac{k}{n})^{2025}=\int_0^1 x^{2025}dx=\frac{1}{2026}\left[x^{2026}\right]_0^1=\frac{1}{2026}.$$

Hence, $\lim\limits_{n\to\infty}\frac{1}{n^{N+1}}\sum\limits_{k=1}^{n}k^N=\frac{1}{N+1}\ \forall N\in\mathbb{N}$.